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IDOL: IoT based intrusion detection using Conv- BiLSTM

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ABSTRACT

Internet of Things (IoT) networks have produced copious amounts of data that can be used to identify intrusions. Network Intrusion Detection Systems are crucial for spotting attacks before they cause any damage as part of a layered defence strategy. In this paper, a novel Intrusion Detection using Optimized deep Learning (IDOL) has been proposed to detect the intrusions like DDoS attacks, Botnet attacks, Convergence attacks, and Ransomware attacks. Initially the IXIA PerfectStorm tool display data with malicious attack from the Network Environment and the data are converted in the data Pre-Processing method. Then the needed data are extracted in the feature extraction process by using BoW technique. From the Extracted data the Feature selection process will select the essential data using Ant Colony Optimization. Finally, the Conv-BiLSTM model will classify the attack into four classes namely DDoS attack, Botnet attack, Convergence attack, and Ransomware attack. The proposed IDOL method achieves the highest accuracy rate of 98.93%. Using the UNSW-NB15 and NSL-KDD datasets, the suggested model achieves an accuracy rate of 98.92% and 98.94% which is greater than the existing techniques such as ML-IMID, IDS-SLoEL, and POS-LightGRM respectively.

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1. Introduction

The IoT is a relatively new technology that links objects to the Internet to improve and support people's lives, careers, and cultures (Sahai, Goel, and Singh 2021). It is a commonly utilised technology in automated network systems. The next wave of Information Technology (IT) advancement and interconnection in the IoT, ranges from Domestic applications to tiny toys to smart cities (Nimbalkar and Kshirsagar 2021). IoT devices work in many settings to achieve a range of objectives; this is reflected in the variety of computer systems and communication tools employed in business, healthcare, education, and agriculture. It significantly impacts nearly every industry in contemporary civilisation, including agriculture, medicine, surveillance, and so on (Saba et al. 2022). IoT uses include implanted blood pressure monitors, pets with transponders, and cars with sensors that warn the driver when the tire pressure is low, which includes any object on Earth that is permitted to receive an IP address and has the capacity to transmit data over a network (Sadikin, Van Deursen, and Kumar 2020).

IDS are the most widely used method for finding intrusions in digital and networked systems (Qiu et al. 2020). Intrusion detection uses two different kinds of detection techniques: anomaly based and signature-

based (Roy et al. 2022). The anomaly-based IDS continuously scans network traffic for changes to the standard network profile and the signature-oriented IDS compares attack patterns already recorded in its database with network traffic patterns (Lazzarini, Tianfield, and Charissis 2023; Verma and Ranga 2020). The majority of intrusion attack problems in the IoT architecture occur at the network, application, and perception layers (Mishra and Pandya 2021).

The intrusion detection systems frequently use a multipronged approach to security, combining several different elements to ensure comprehensive threat detection. In 2025, 75.44 billion devices will be linked to the IoT, as seen in Figure 1, producing 79 zettabytes (ZB) of data. One essential element of digitisation for a changing society is acknowledged as the IoT.

IoT networks require a different approach to security than traditional networks because of their unique characteristics, which include heterogeneity, resource constraints, and a broad attack surface. The inadequacies of current security standards are highlighted by the growing attraction of connected devices to hackers seeking to exploit security flaws and obtain private information without authorisation. The suggested method's primary contributions are as follows:

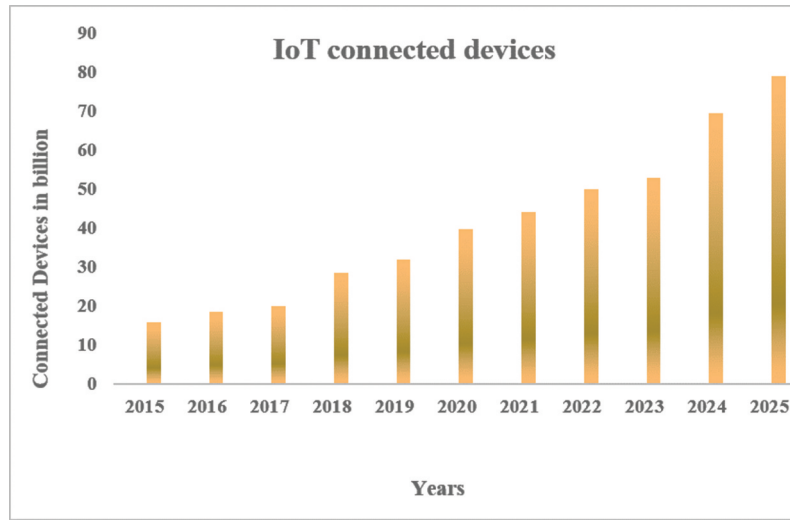


Figure 1. Growth of IoT devices from 2015 to 2025.

- Initially the IXIA PerfectStorm tool display data with malicious attack from the Network Environment and the data are converted in the data Pre-Processing method.
- Then the needed data are extracted in the feature extraction process by using BoW technique.
- From the Extracted data the Feature selection process will select the essential data using Ant Colony Optimization.
- Finally, the Conv-BiLSTM model will divide the attack into four different types such as ransomware, convergent, botnet, and DDoS attacks.

This study's remaining sections are explained as follows: The research is analysed in Section II using the literature as a guide. The proposed strategy is thoroughly explained in Section III. While the outcome and discussion are found in Section IV, the conclusion is found in Section V.

2. Literature review

In (De Souza et al. 2020) proposed the DNN-KNN hybrid binary classification technique. This method is used to choose qualities according to the speed of information acquisition. For the CICIDS2017 dataset with the NSL-KDD dataset, the method described in this resource produced 99.85% and 99.77% accuracy rates, respectively. The primary disadvantage of the proposed method is low accuracy for multi-class detection and only a few experiments are performed for evaluation.

In (Liu et al. 2021) developed a particle swarm optimisation (PSO-LightGBM) gradient descent technique for detecting intrusions. This method identifies and categorises hazardous data by using PSO-LightGBM to extract features from the data, which is

then put into one-class SVM (OCSVM). The proposed PSO-LightGBM achieves an accuracy rate of 86.68%. The low detection rate and restricted scalability of conventional intrusion detection systems are major drawbacks.

In (Gad, Nashat, and Barkat 2021) suggested Vehicular ad hoc networks (VANETs) that make it possible for automobiles to connect through the network of wireless communications. This method's objective is to high degree of security capabilities by using IDS. This method identifies a top accuracy rate of 0.980%. The drawback of this approach is that some elements that are removed can be essential for accurately portraying the class.

In (Keserwan et al. 2021) proposed a GWO – PSO – RF NIDS model that recognises several types of assaults on IoT networks. In these more frequent records raise the DR for attack classes that are minorities. Compared to other contemporary methods, the suggested model had a high accuracy, averaging 99.66% across the entire datasets. The main drawbacks of this approach are its slow convergence, low precision, and poor local minimum searching capabilities.

In (Oseni et al. 2022) proposed an explainable IDS for IoT networks based on deep learning that improves the DL-based IDS's resilience and slide. The approach generates its results using a deep learning-based IDS and SHAP apparatus. The testing results show that the suggested framework performs exceptionally well and can protect the networks from sophisticated cyberattacks, with an accuracy of 99.15% and an F1 score of 98.83%.

In (Asif et al. 2022) suggested a MapReduce-based Intelligent Model for Network Intrusion Detection (MR-IMID) that uses several data classification locations to detect network intrusions. This combines real-time utilisation of different network sources for

intrusion detection. Training phase detection accuracy is 95.7% and validation phase detection accuracy is 97.7% for the recommended model. In the future, the suggested approach will be able to detect incursions fast and save the information in the database to detect attacks early.

In (Hazman et al. 2023) suggested MR-IMID a MapReduce-Based Intelligent Model for Intrusion Detection to find intrusions. Based on the performance comparisons, this model is the most reliable, efficient, and least expensive over the long term. In this accuracy, recall, and precision scores are nearly 99.9%. Future research will use deep learning techniques to create an intrusion detection model and multi-class categorisation.

3. Proposed method

In this section a novel Intrusion Detection using Optimized deep Learning (IDOL) have been proposed to detect and classify the attack such as DDoS attack, Botnet attack, Convergence attack and Ransomware attack. The proposed IDOL technique has been shown in Figure 2.

3.1. Data source

The connection that is established from a server to a database is designated as a Data Source. In this the IXIA PerfectStorm system shows the network flow dataset in which regular data and malicious attacks coexisted in an actual network setting. These features fall into seven categories: fundamental, content, time, connection, flow features, labelled, and general-purpose features. In addition, there are three binary features, two timestamp features, six nominal features, ten float features, and twenty-eight integer features.

3.2. Pre-processing

Data preparation can lower the amount of raw data by eliminating unnecessary information unrelated to modelling. This speeds up the model's training process and lowers the dimension of the data. Pre-processing includes various methods for selecting, extracting and scaling features as well as methods for cleaning, converting, and normalising the data. In this study, pre-processing of data mainly includes the data conversion process.

3.2.1. Data conversion

Changing a data format is known as data conversion. It is easier to finish the procedure if the destination format offers the same data structures and features as the source format. If the destination format matches the source data in terms of characteristics and data structures, it can be done easily. This process converts the qualitative data from the original dataset into numerical data.

3.3. Feature extraction

Feature extraction is the process of identifying and picking out the most crucial details or attributes from a collection of data. The process of extracting features involves two steps: feature development and feature selection. The conventional feature design and feature selection techniques have a major influence on an intrusion detection system's efficacy. In this, the most significant features of the data are extracted using the BoW technique.

3.3.1. BoW

Bag Of Words (BoW) is the widely used method for extracting essential data. The system calls monitored in every time frame are built into a bag of document, and the system calls that are executed within each

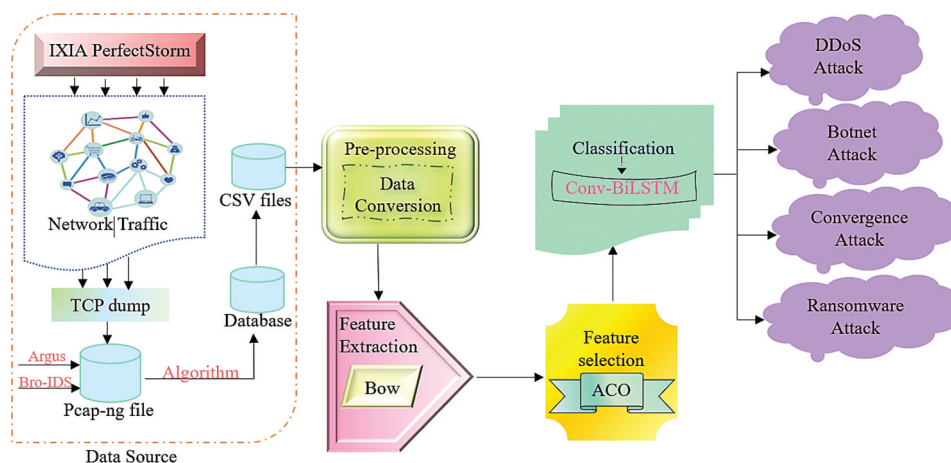


Figure 2. Block diagram of proposed IDOL model.

time window are then created into a vector. Each vector location represents n system calls, and depending on the chosen n-Gram, each bag entry may contain one or more system calls. Finally, there are two methods for filling vector locations. First, there is a measure called frequency that keeps track of how many times each n -gram happens within the given time frame. TF-IDF is the next one, where considers the number of different systems calls per time window and the repetitions to determine the significance of the n-Gram in the vector. Equation 1 determines a term's TF-IDF value in a document d that is a part of the vector V , N denotes the quantity of documents, n was the quantity of terms in d and $f_{a,d}$ is the frequency of term a in d .

$$l = \sum_{r=1}^R \frac{1}{R} l_r \quad (1)$$

At last, global model is distributed to each data set. It is used to analyse data once the model converges after several iterations.

3.4. Feature selection

Feature selection is the process of choosing the most trustworthy, relevant, and unique features to be used in the model-building process. Through removing superfluous data, simplifying data, increasing machine learning method detection rates, and reducing false alarms, feature selection facilitates calculations. Several techniques have been used in this field of study to create lightweight IDSs. In this feature selection process is done by applying the Ant Colony Optimization method.

3.4.1. Ant colony optimization (ACO)

ACO is a probabilistic method for resolving computer issues that boil down to identifying optimal pathways across graphs. The ant system (AS) was an acronym given to the first computational model. An alternative approach to stochastic combinatorial optimisation is this one. The search operations have been divided among: 'ants' – agents with innate, fundamental tendencies that mimic the actions of actual ants. The primary goal was to employ artificial ant colonies as an optimisation tool rather than to replicate ant colonies. As an outcome, the technique shows multiple variations that match the actual (natural) ant colony: Artificial ants live in a world with distinct time zones, have some memory, and are not blind. ACO is particularly effective at managing high-dimensional data because it balances exploration and exploitation by effectively identifying the best feature subsets by imitating ant foraging behaviour. This can improve detection accuracy and processing performance by

drastically reducing the dataset's dimensionality. ACO is a good fit for intrusion detection feature relevance may change over time.

An evaluation function for a single subset, like the Fisher Discrimination Rate (FDR), an entropy-based metric, or a crude set dependency measure, could be an appropriate heuristic desirability of traversing across features. The probabilistic transition rule can be obtained by combining edge pheromone levels and traversal have heuristic attraction, which shows the probability that an ant will decide to visit a feature b at time t :

$$H_{ab}^f(t) = \begin{cases} \frac{[\tau_{ab}(t)]^\alpha \cdot [\eta_{ab}]^\beta}{\sum_{p \in B_a^f} [\tau_{ap}(t)]^\alpha \cdot [\eta_{ap}]^\beta} & \text{if } a \in B^f \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Where f is the number of ants, B_a^f is the collection of nodes a neighbours that the ant f has not yet visited, η_{ab} is a feature's heuristic desirability b while in the feature a . Two parameters define the pheromone value and the heuristic information, $\alpha > 0$ and $\beta > 0$.

These explore edges in a probabilistic manner starting from these starting points and continue until a traversal halting condition is met. Subsets that are thus produced are subsequently assembled and assessed. The cycle repeats when a fresh batch of ants is produced, the pheromone is updated, and one of the two possibilities is realised.

The following formula is used to update the pheromone on each edge:

$$\tau_{ab}(t+1) = (1 - \rho) \cdot \tau_{ab}(t) + \rho \cdot \Delta\tau_{ab}(t) \quad (3)$$

Where:

$$\Delta\tau_{ab}(t) = \sum_{f=1}^n (\gamma(R^f) / |R^f|) \quad (4)$$

This is valid if one has travelled across the edge $(a; b)$, $\Delta\tau_{ab}(t)$ is otherwise 0. Deterioration Constant, ρ is used to mimic the pheromone's evaporation and has a value of $0 \leq \rho \leq 1$ and R^f is the feature subset found by ant f . Based on these standards, all ants update the pheromone.

3.5. Conv-BiLSTM model

Through Conv-BiLSTM model the attack from the data source is classified. The following section covers the suggested model's structure. The bidirectional LSTM layers, that are essential for comprehending intricate attack patterns over time, capture sequential dependencies in both forward and backward directions, while the convolutional layers probably aid in the extraction of local patterns from the input data, identifying important features pertinent to attack detection. It would be better to highlight the architecture's contributions with a more thorough explanation

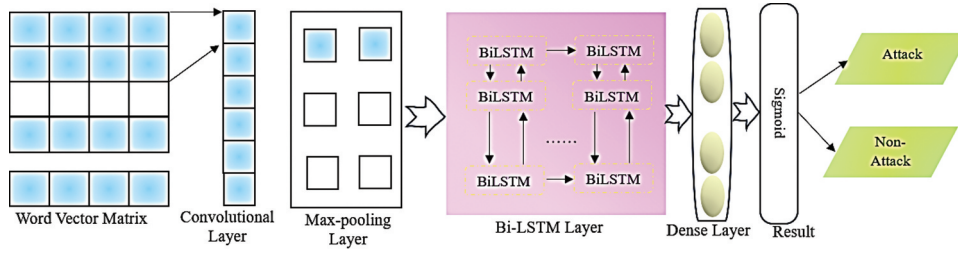


Figure 3. Architecture of conv-BiLSTM model.

that includes its design reasoning and function in enhancing categorisation outcomes. The five phases that comprise the high-level structure of the Conv-BiLSTM model are illustrated in Figure 3.

Combining CNN and BiLSTM enables IDS to manage both geographical and temporal aspects of network traffic data by utilising their complementing strengths. CNN is especially effective at detecting localised patterns and extracting spatial features from structured data, including harmful signatures and anomalies in packet headers. However, by processing information in both directions BiLSTM is able to recognise contextual patterns and long-term dependencies across network events, as well as temporal dependencies and sequential linkages in data.

3.5.1. Word vectorization

The outcome of the feature selection procedure is eventually divided by the network into distinct words or tokens. This will convert every token into a vector's numerical value. To build the word vector matrix, GloVe and Word2Vec are examples of pretrained word embedding models. The model's accuracy is assessed using Word2Vec as well as GloVe models. The input text is determined as $T = \{e_1, e_2, \dots, e_n\}$ where T specifies the length n and e_1 across e_n are word vectors of length f .

$$T = \{e_1, e_2, \dots, e_n\} \in S^{n \times f} \quad (5)$$

Every output text ought to have the same length (l), as the input texts may differ in length. Its length was not extended with any cushioning. Any text that exceeds the allotted length, l , will be terminated. However, no additional paces are added to the total if the text is shorter than length (l). All texts have a uniform matrix dimension as a result. The definitions for any $-$ dimensional text are as follows:

$$T = \{e_1, e_2, \dots, e_n\} \in R^{l \times f} \quad (6)$$

3.5.2. Convolutional layer

The convolution layer, the basic stage of a CNN model is excellent at determining which words in a phrase are most important.

The feature maps generate that the filter F_n are shown, where $1 \leq n \leq N$:

$$a_i^n = f(e^n \otimes Y_{i:i+e-1} + p^n) \quad (7)$$

The word vector dimension d , the convolution process, and b^n the bias of the filter F_n are represented by the weight matrix $e \in R^{q \times f}$. The filter is indicated as $X_{i:i+q-1}$. F_n extract feature $X_{i:i+q-1}$ from X_i . The RELU function was employed in this investigation to achieve non-linear activation. The feature maps for the sentence of length l were made as follows:

$$m = m_1, m_2, \dots, m_i, m_l \quad (8)$$

3.5.3. Max-pooling layer

The pooling layer makes use of the best attributes $\hat{m} = \max\{m\}$ to calculate the local adequate statistics after the convolution procedure has produced the feature maps.

3.5.4. Bi-lstm layer

When compared to LSTM, Bi-LSTM uses hidden states to enable both forward and backward flow of information. It could improve Bi-LSTMs' capacity to identify contextual cues. Instead of requiring decay to integrate future information, as is the case with the typical RNN model, it is possible to retain input data from the past as well as the future by utilising these bidirectional links. The results obtained from two LSTM networks operating in opposition to one another are typically connected to create a Bi-LSTM. Here the Bi-LSTM will examine the weight of the result.

3.5.5. Dense layer and result

In this model, a dense layer connects the inputs and outputs using values. The final output is generated in the final layer using a sigmoid function. The randomly generated average is transformed into binary form. The predictive performance of the sigmoid function is shown by the equation (9).

$$R = \text{Sigmoid}(wf + p) \quad (9)$$

Based on the sentiment analysis result, binary cross-entropy can be used to classify the data as either

0 or 1. A score of 0 in this analysis denotes an unfavourable perspective, while a score of 1 denotes a favourable viewpoint.

4. Result and discussion

This section presents the evaluation results of the attack categorisation on an IoT platform. Performance is examined using a number of evaluation criteria. The attack is found on a Windows computer running Python with libraries on an Intel Core i7 processor with 16 GB of RAM. The effectiveness of the proposed approach is assessed using the dataset that has been provided.

4.1. Description of the dataset

4.1.1. NSL-KDD dataset (Dhanabal and Shantharajah 2015)

An enhanced version of the KDD'99 protocol is the NSL-KDD data gathering. Researchers testing different intrusion detection methods can benefit greatly from this benchmark data collection. It functions as a standard dataset for assessing intrusion detection systems. The possibility of biased learning outcomes is decreased by NSL-KDD's more balanced distribution of attack and regular traffic data. NSL-KDD has transformed into a dependable and popular dataset for creating and evaluating IDS models according to these enhancements.

4.1.2. UNSW-NB15 dataset (Moustafa and Slay 2016)

UNSW-NB15 is one dataset pertaining to network intrusions. It includes nine distinct attack types, including as viruses, fuzzes, backdoors, and denial-of-service assaults. Raw network packets are included in the collection. The IXIA PerfectStorm program was used to produce the raw network packets. The dataset allows for the evaluation of machine learning and deep learning models for intrusion detection because it contains 49 features that were extracted from raw network packets. Studying and improving cybersecurity systems is made easier by its extensive feature set and realistic attack scenarios."

4.2. Description of evaluation matrices

4.2.1. The evaluation measures are F1-score, accuracy, recall, and precision. These measures are influenced by four variables

- True Positive (Tr. Po): It displays the number of malicious observations on IoT devices that the model correctly classified as malicious activity.

- True Negative (Tr. Ne): This illustrates the percentage of observations of common IoT devices that the model correctly identified as typical behaviour.
- False Positive (Fa. Pe): This demonstrates the number of valid observations of IoT devices whose behaviour was incorrectly classified by the model as hostile.
- False Negative (Fa. Ne): This demonstrates the number of negative evaluations on Internet of Things devices that the model mistook for typical behaviour.

4.3. Performance evaluation

Based on the qualities mentioned, a number of metrics are computed to assess IDS performances.

The rate of correctly classifying a category is measured by accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Precision quantifies the percentage of true positives among all alerts, or the dependability of your intrusion detection system when it sounds an alarm.

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

Sensitivity or TPR are other names for recall, which is determined by applying equation, is related to the likelihood of finding intrusions.

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

The combined REC and PREC of the recommended OPTIMIST is shown to be generally effective by the F1-Score.

$$F1 - Score = 2 \times \frac{Precision * Recall}{Precision + Recall} \quad (13)$$

The performance comparison of F1 score, recall, accuracy, and precision is displayed in Figure 4.

4.4. Comparative analysis

The comparison of accuracy, F1-score, precision, and recall utilising the UNSW-NB15 and NSL-KDD datasets is displayed in Figure 5. The proposed approach attains accuracy rates of 98.92% and 98.94%, respectively, using the NSL-KDD and UNSW-NB15 datasets.

Figure 6(a,b) show the training and test data sets along with the accuracy and loss curves. The model's accuracy improves over training epochs on both the training and testing sets, as seen by the Accuracy Curve in Subfigure (a). Training and validation losses tend to decline, as indicated by the Loss Curve in

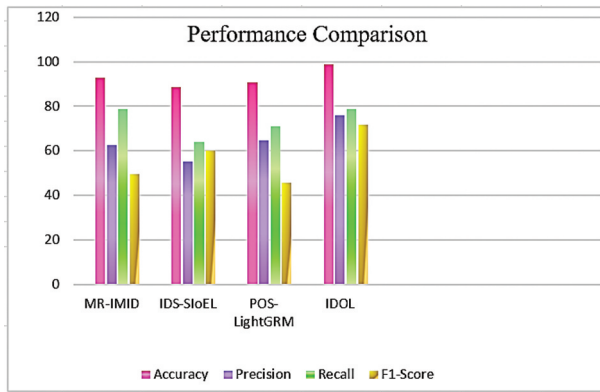


Figure 4. Performance comparison.

Subfigure (b). These curves provide useful insights into the learning process to guarantee optimal intrusion detection model performance.

4.5. Security rate

The security rates of proposed and accessible intrusion detection techniques are contrasted in Figure 7. The

picture shows the effectiveness that each tactic is in protecting systems. Potential improvements in intrusion detection technology are illustrated graphically in this paper.

4.6. Detection rate

A measure of coverage that assesses the attackers' capacity to be recognised is detection rate (DR), which expresses the accurate percentage of all positive instances. The proposed method's detection rate is contrasted with that of several other techniques, such as ML-IMID, IDS-SLoEL, and POS-LightGRM, using the dataset in Figure 8. Among all possible targets, this assesses a process or system's capacity to identify and locate a desired target. Compared to other systems now in use, the suggested intrusion detection technology has a high detection rate.

4.7. Time efficiency

The time efficiency of the suggested IDOL is shown in Figure 9 compared to other current techniques

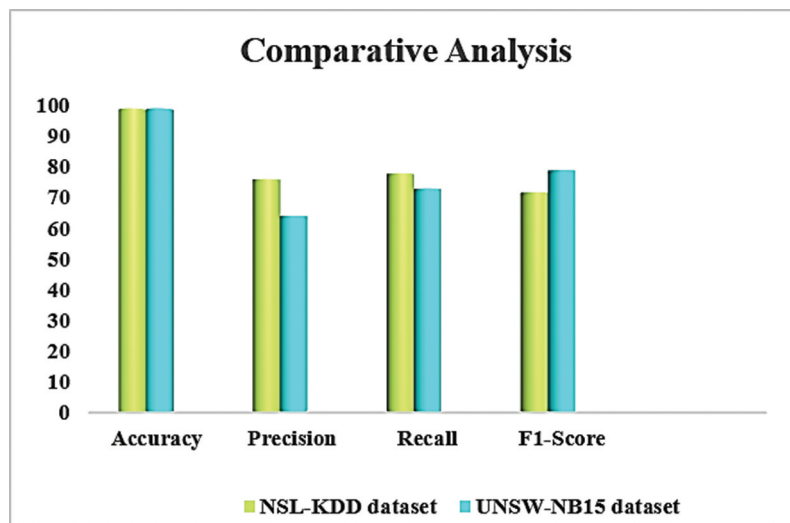


Figure 5. Comparative analysis with dataset.

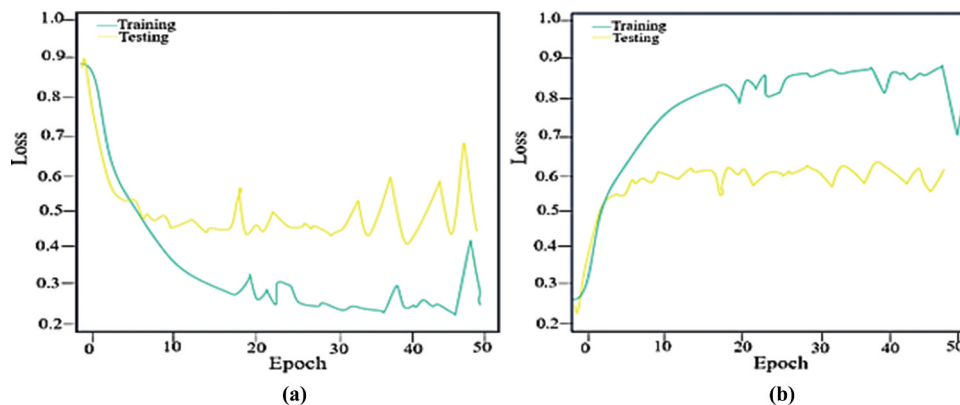


Figure 6. (a) Accuracy curve, (b) loss curve.

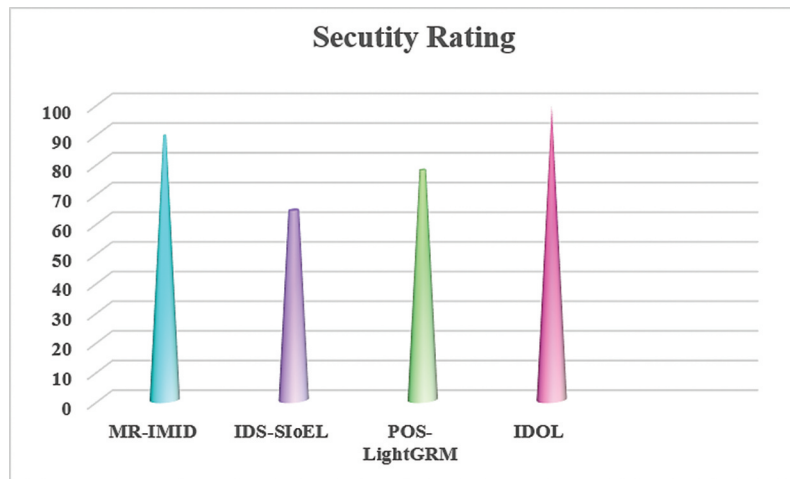


Figure 7. Comparison of security rates.

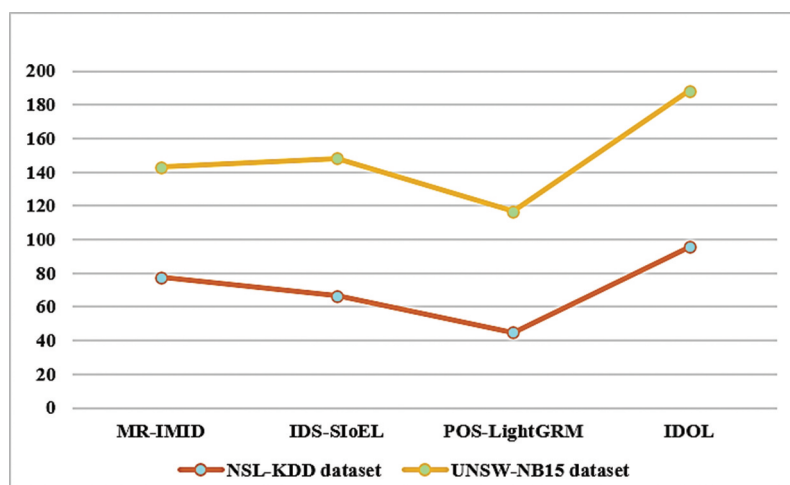


Figure 8. Detection rate comparison.

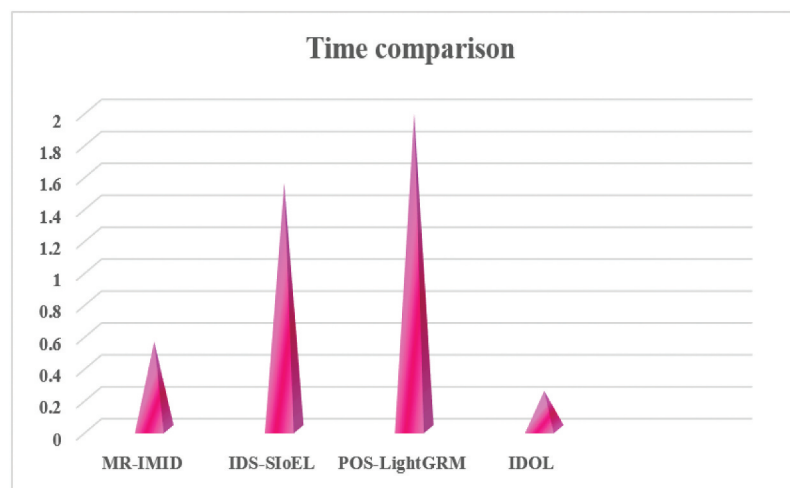


Figure 9. Comparison of time.

such as IDS-SLoEL, MR-IMID, and POS-LightGRM. Time efficiency in vulnerability detection refers to how quickly and efficiently vulnerabilities in a system are identified and analysed. Compared to

the MR-IMID, IDS-SLoEL, and POS-LightGRM approaches, which require 0.55 milliseconds, 1.54 milliseconds, and 1.98 milliseconds, respectively, the proposed system's time of 0.24 milliseconds is

comparatively fast. It illustrates that the suggested IDOL technique finds vulnerabilities faster than the existing approaches.

5. Conclusion

In this paper a novel Intrusion Detection using Optimized deep Learning (IDOL) have been proposed to detect the intrusions like DDoS attack, Botnet attack, Convergence attack, and Ransomware attack. In this data conversion is done in pre-processing method and from the converted data the characteristic is extracted using the BoW technique. Then the essential features are selected after the Ant Colony Optimization process. Finally, the Conv-BiLSTM model will classify the attack into four classes namely DDoS attack, Botnet attack, Convergence attack, and Ransomware attack. Using the NSL-KDD and UNSW-NB15 dataset the efficiency of the proposed system is evaluated. With the NSL-KDD dataset and the UNSW-NB15 dataset, the suggested model achieves an accuracy rate of 98.92% and 98.94% which is greater than the existing techniques such as ML-IMID, IDS-SLoEL, and POS-LightGRM. Future research will focus on enhancing the model's awareness of the most recent security concerns by including real-time threat information sources.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions statement

The authors confirm contribution to the paper as follows: Study conception and design: Vijayalakshmi. C, Padmajothi. V, Suguna Devi. R; Data collection: Priscilla whitin, Rajakumar. S, Subraja Rajaretnam; Analysis and interpretation of results: Vijayalakshmi. C, Padmajothi. V, Suguna Devi. R; Draft manuscript preparation: Priscilla whitin, Rajakumar. S, Subraja Rajaretnam. All authors reviewed the results and approved the final version of the manuscript.

Availability of data and material

Data sharing is not applicable to this article as no new data were created or analysed in this Research.

Human and animal rights

This article does not contain any studies with human or animal subjects performed by any of the authors.

Informed consent

I certify that I have explained the nature and purpose of this study to the above-named individual, and I have discussed the potential benefits of this study participation. The questions the individual had about this study have been answered, and we will always be available to address future questions.

Ethical approval

My research guide reviewed and ethically approved this manuscript for publishing in this Journal.

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